



PQ

Power Query Data Cleaning

HR Employee Dataset · Before & After

*From messy, inconsistent records to a clean,
analysis-ready table - 22 applied steps in Power Query.*

The Messy Data

Raw HR export before cleaning

	A	B	C	D	E	F	G	H	I
1	Employee_ID	Full_Name	Email	Phone	Join_Date	Department	City	Salary	
2	483	Daniel Brown	Daniel.Brown@company			Operations	New York	76342.26	
3									
4	101	Smith, Sophia	sophia.smith@gmail.com		2024.04.06	Finance	Los Angeles	104278.02	
5	73	emma LEE	emma.lee@gmail.com	643-622-4176	08/30/2023	Marketing	Chicago	\$47754.46	
6	494	Emma Smith	emmasmith@yahoo.com			Sales	Los Angeles		
7	375	Ava Smith		3899580938	2024.05.23	HR	Seattle		
8	374	Emma Wilson	emma.wilson@gmail.com	(989) 412-2875		Sales	Seattle		
9	237	daniel CLARK	daniel.clark@gmail.com		09-04-2023	Marketing	Los Angeles	72828.23	
10	403	david LEE	davidlee@yahoo.com		04/30/2023		Phoenix		
11	304	David Taylor	davidtaylor@yahoo.com	3882455823	02-06-2023	HR		54627.32	
12	222	James Hall	james.hall@gmail.com	985-801-7950		Sales	Seattle	\$114577.6	
13	280	ava WALKER		(145) 937-4825	06-09-2023	Operations	Los Angeles	108187.71	
14	414	Michael Smith	michael.smith@gmail.com	(446) 422-7580	07-11-2024		Dallas		
15	475	Emma Smith	Emma.Smith@company	9333460559			New York	57702.97	
16	34	sarah ALLEN	sarah.allen@gmail.com		2024.03.14		Seattle	36206.11	
17	133	Daniel Johnson	danieljohnson@yahoo.com		2023.11.21		Miami		
18	406	Ava Clark				HR	New York		
19	125	Lee, Sophia	sophia.lee@gmail.com	859-249-1265	2023.07.20			\$81281.06	
20	284	Daniel Smith	danielsmith@yahoo.com		25-11-2024	IT			
21	120	daniel WALKER	Daniel.Walker@company		2024.09.04		Phoenix	\$108863.25	
22	411	Ava Johnson		(918) 568-2817	06/21/2023	IT	Seattle	\$111988.72	
23									
24	92	david WALKER	David.Walker@company			Operations	Los Angeles		
25	60	James Lee		968-943-8450	10/19/2024	HR	Los Angeles	\$60015.82	
26									
27	161	sarah SMITH	sarah.smith@gmail.com	635-629-5247	10/03/2023	IT			
28	201	sophia TAYLOR			31-05-2024		Phoenix	41084.58	
29	108	david WALKER	davidwalker@yahoo.com		23-08-2024	HR	New York	\$90285.06	
30	454	Taylor, Emma	emmataylor@yahoo.com	839-933-8559		Operations	Chicago	48862.63	
31	45	Brown, John	John.Brown@company	5686850445	2024.02.23	Operations	Dallas	\$39413.99	
32	107	Lee, Olivia		9504724761	2023.05.30	Operations	Seattle	33785.98	
33	232	emma SMITH	Emma.Smith@company	(910) 469-7358	2024.06.17		Dallas	\$53698.45	

Inconsistent name casing, missing emails/phones, mixed date formats (2024.04.06 vs 08/30/2023), and blank department, city, or salary fields.

After Cleaning

Standardized, complete, and analysis-ready

	A	B	C	D	E	F	G	H
1	Employee_ID	Full Name	Email	Phone	Join Date	Department	City	Salary
2	483	Daniel Brown	daniel.brown@company	Not Found	Not Found	Operations	New York	\$76,342
3	101	Smith, Sophia	sophia.smith@gmail.com	Not Found	2024/04/06	Finance	Los Angeles	\$104,278
4	73	Emma Lee	emma.lee@gmail.com	6436224176	08/30/2023	Marketing	Chicago	\$47,754
5	494	Emma Smith	emmasmith@yahoo.com	Not Found	Not Found	Sales	Los Angeles	\$0
6	375	Ava Smith	Not Found	3899580938	2024/05/23	HR	Seattle	\$0
7	374	Emma Wilson	emma.wilson@gmail.com	989 4122875	Not Found	Sales	Seattle	\$0
8	237	Daniel Clark	daniel.clark@gmail.com	Not Found	09/04/2023	Marketing	Los Angeles	\$72,828
9	403	David Lee	davidlee@yahoo.com	Not Found	04/30/2023	Not Found	Phoenix	\$0
10	304	David Taylor	davidtaylor@yahoo.com	3882455823	02/06/2023	HR	Not Found	\$54,627
11	222	James Hall	james.hall@gmail.com	9858017950	Not Found	Sales	Seattle	\$114,578
12	280	Ava Walker	Not Found	145 9374825	06/09/2023	Operations	Los Angeles	\$108,188
13	414	Michael Smith	michael.smith@gmail.com	446 4227580	07/11/2024	Not Found	Dallas	\$0
14	475	Emma Smith	emma.smith@company	9333460559	Not Found	Not Found	New York	\$57,703
15	34	Sarah Allen	sarah.allen@gmail.com	Not Found	2024/03/14	Not Found	Seattle	\$36,206
16	133	Daniel Johnson	danieljohnson@yahoo.com	Not Found	2023/11/21	Not Found	Miami	\$0
17	406	Ava Clark	Not Found	Not Found	Not Found	HR	New York	\$0
18	125	Lee, Sophia	sophia.lee@gmail.com	8592491265	2023/07/20	Not Found	Not Found	\$81,281
19	284	Daniel Smith	danielsmith@yahoo.com	Not Found	25/11/2024	IT	Not Found	\$0
20	120	Daniel Walker	daniel.walker@company	Not Found	2024/09/04	Not Found	Phoenix	\$108,863
21	411	Ava Johnson	Not Found	918 5682817	06/21/2023	IT	Seattle	\$111,989
22	92	David Walker	david.walker@company	Not Found	Not Found	Operations	Los Angeles	\$0
23	60	James Lee	Not Found	9689438450	10/19/2024	HR	Los Angeles	\$60,016
24	161	Sarah Smith	sarah.smith@gmail.com	6356295247	10/03/2023	IT	Not Found	\$0
25	201	Sophia Taylor	Not Found	Not Found	31/05/2024	Not Found	Phoenix	\$41,085
26	108	David Walker	davidwalker@yahoo.com	Not Found	23/08/2024	HR	New York	\$90,285
27	454	Taylor, Emma	emmataylor@yahoo.com	8399338559	Not Found	Operations	Chicago	\$48,863
28	45	Brown, John	john.brown@company	5686850445	2024/02/23	Operations	Dallas	\$39,414
29	107	Lee, Olivia	Not Found	9504724761	2023/05/30	Operations	Seattle	\$33,786
30	232	Emma Smith	emma.smith@company	910 4697358	2024/06/17	Not Found	Dallas	\$53,698
31	214	Sarah Clark	sarah.clark@company	Not Found	17/05/2023	Operations	Not Found	\$0
32	196	Michael Wilson	Not Found	231 8585237	2023/08/16	Operations	Miami	\$0
33	148	Taylor, James	james.taylor@gmail.com	7226553010	Not Found	HR	Not Found	\$40,565

Names capitalized, emails standardized, phone numbers formatted, dates unified as YYYY/MM/DD, and nulls replaced with “Not Found” or \$0.

Behind the Scenes

Power Query Editor - Applied Steps panel

The screenshot displays the Power Query Editor window for a query named 'Messy_Data'. The ribbon includes tabs for File, Home, Transform, Add Column, and View. The 'Transform' tab is active, showing various data transformation options like 'Data Type: Whole Number', 'Merge Queries', 'Append Queries', 'Combine Files', 'Manage Parameters', 'Data source settings', 'New Source', and 'Recent Sources'. The main area shows a table with columns: Employee_ID, Full_Name, Email, Phone, Join_Date, Department, and City. The table contains 25 rows of data. On the right, the 'Query Settings' pane is open, showing the 'APPLIED STEPS' section with a list of 22 steps recorded automatically. The steps include: Navigation, Promoted Headers, Changed Type, Removed Blank Rows, Removed Duplicates, Trimmed Text in Full_Name, Capitalized Each Word in F..., Trimmed Text in Email Col..., Lowercased Text in Email C..., Replace Null with Not Fou..., Removed Hyphens in Phone, Removed Opening Parenth..., Removed Closing Parenthe..., Trimmed Phone Numbers, Replace Null with Not Fou..., Replace ' ' with ' / ' in Join_D..., Replace ' ' with ' / ' in Join..., Replace null with Not Foun..., Replacenull with Not Foun..., Replacenull with Not Foun..., and Replace null to 9 in Salary. The last step, 'Changed Salary type in Cur...', is highlighted in green.

Queries [1]
Messy_Data

Formula Bar: `= Table.TransformColumnTypes(#"Replace null to 9 in Salary",{{"Salary", Currency.Type}})`

	Employee_ID	Full_Name	Email	Phone	Join_Date	Department	City
1	483	Daniel Brown	daniel.brown@company	Not Found	Not Found	Operations	New York
2	101	Smith, Sophia	sophia.smith@gmail.com	Not Found	2024/04/06	Finance	Los Angeles
3	73	Emma Lee	emma.lee@gmail.com	6436224176	08/30/2023	Marketing	Chicago
4	494	Emma Smith	emmasmith@yahoo.com	Not Found	Not Found	Sales	Los Angeles
5	375	Ava Smith	Not Found	3899580938	2024/05/23	HR	Seattle
6	374	Emma Wilson	emma.wilson@gmail.com	989 4122875	Not Found	Sales	Seattle
7	237	Daniel Clark	daniel.clark@gmail.com	Not Found	09/04/2023	Marketing	Los Angeles
8	403	David Lee	davidlee@yahoo.com	Not Found	04/30/2023	Not Found	Phoenix
9	304	David Taylor	davidtaylor@yahoo.com	3882455823	02/06/2023	HR	Not Found
10	222	James Hall	james.hall@gmail.com	9858017950	Not Found	Sales	Seattle
11	280	Ava Walker	Not Found	145 9374825	06/09/2023	Operations	Los Angeles
12	414	Michael Smith	michael.smith@gmail.com	446 4227580	07/11/2024	Not Found	Dallas
13	475	Emma Smith	emma.smith@company	9333460559	Not Found	Not Found	New York
14	34	Sarah Allen	sarah.allen@gmail.com	Not Found	2024/03/14	Not Found	Seattle
15	133	Daniel Johnson	danieljohnson@yahoo.com	Not Found	2023/11/21	Not Found	Miami
16	406	Ava Clark	Not Found	Not Found	Not Found	HR	New York
17	125	Lee, Sophia	sophia.lee@gmail.com	8592491265	2023/07/20	Not Found	Not Found
18	284	Daniel Smith	danielsmith@yahoo.com	Not Found	25/11/2024	IT	Not Found
19	120	Daniel Walker	daniel.walker@company	Not Found	2024/09/04	Not Found	Phoenix
20	411	Ava Johnson	Not Found	918 5682817	06/21/2023	IT	Seattle
21	92	David Walker	david.walker@company	Not Found	Not Found	Operations	Los Angeles
22	60	James Lee	Not Found	9689438450	10/19/2024	HR	Los Angeles
23	161	Sarah Smith	sarah.smith@gmail.com	6356295247	10/03/2023	IT	Not Found
24	201	Sophia Taylor	Not Found	Not Found	31/05/2024	Not Found	Phoenix
25	400	David Wilson	david.wilson@company	Not Found	2024/03/14	Not Found	Seattle

Query Settings

PROPERTIES
Name
Messy_Data

APPLIED STEPS

- Navigation
- Promoted Headers
- Changed Type
- Removed Blank Rows
- Removed Duplicates
- Trimmed Text in Full_Name
- Capitalized Each Word in F...
- Trimmed Text in Email Col...
- Lowercased Text in Email C...
- Replace Null with Not Fou...
- Removed Hyphens in Phone
- Removed Opening Parenth...
- Removed Closing Parenthe...
- Trimmed Phone Numbers
- Replace Null with Not Fou...
- Replace ' ' with ' / ' in Join_D...
- Replace ' ' with ' / ' in Join...
- Replace null with Not Foun...
- Replacenull with Not Foun...
- Replacenull with Not Foun...
- Replace null to 9 in Salary
- Changed Salary type in Cur...

22 steps recorded automatically - every transformation is repeatable with one click of Refresh.

What Each Step Did

Grouping the 22 applied steps by purpose

1

Text Cleanup

Trimmed extra spaces, capitalized each word in Full_Name, and lowercased Email addresses.

2

Fixing Missing Values

Replaced blank Phone, Department, City fields with “Not Found”, and blank Salary with 9.

3

Formatting Consistency

Removed hyphens and parentheses from Phone numbers; standardized Join_Date to one format.

4

Data Quality

Removed blank rows and duplicate records, then set Salary to Currency type.

Key Takeaway

**Clean data isn't deleted -
it's corrected, standardized,
and made trustworthy.**

- Practiced 22 Power Query steps on one dataset
- Focused on text, dates, phone numbers & currency
- Built a repeatable, refreshable cleaning pipeline